

Code:

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#include <stdio.h>
#include <stdlib.h>

#define MAXN 100005
#define MAXM 200005

int head[MAXN], to[MAXM], nxt[MAXM], edge_cnt;
int head_rev[MAXN], to_rev[MAXM], nxt_rev[MAXM], edge_rev_cnt;
int vis[MAXN];

void add_edge(int u, int v) {
    to[++edge_cnt] = v;
    nxt[edge_cnt] = head[u];
    head[u] = edge_cnt;

    to_rev[++edge_rev_cnt] = u;
    nxt_rev[edge_rev_cnt] = head_rev[v];
    head_rev[v] = edge_rev_cnt;
}

void dfs1(int u) {
    vis[u] = 1;
    for (int e = head[u]; e; e = nxt[e]) {
        int v = to[e];
        if (!vis[v]) dfs1(v);
    }
}

void dfs2(int u) {
    vis[u] = 1;
    for (int e = head_rev[u]; e; e = nxt_rev[e]) {
        int v = to_rev[e];
        if (!vis[v]) dfs2(v);
    }
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;
```

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int orig_m = m;
while (m--) {
    int u, v;
    scanf("%d %d", &u, &v);
    add_edge(u, v);
}

// Step 1: Reachability from City 1
for (int i = 1; i <= n; i++) vis[i] = 0;
dfs1(1);
for (int i = 1; i <= n; i++) {
    if (!vis[i]) {
        printf("1\n%d 1\n", i);
        return 0;
    }
}

// Step 2: Reachability to City 1
for (int i = 1; i <= n; i++) vis[i] = 0;
dfs2(1);
for (int i = 1; i <= n; i++) {
    if (!vis[i]) {
        printf("1\n1 %d\n", i);
        return 0;
    }
}

printf("0\n");
return 0;
}

```

Input:

```

4 5
1 2
2 3
3 1
1 4
4 5

```

Output:

4 5

1 2

2 3

3 1

1 4

4 5